STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – IV

Deflection of Beams

Handwritten Study Notes

Syllabus topics covered in this unit :

Double integration and Macaulay's methods — Slope and deflection for cantilever, simply supported and overhanging beams under point loads, U.D.L., U.V.L., partial U.D.L., couple and combinations — Mohr's theorems — Moment area method — application to cantilevers

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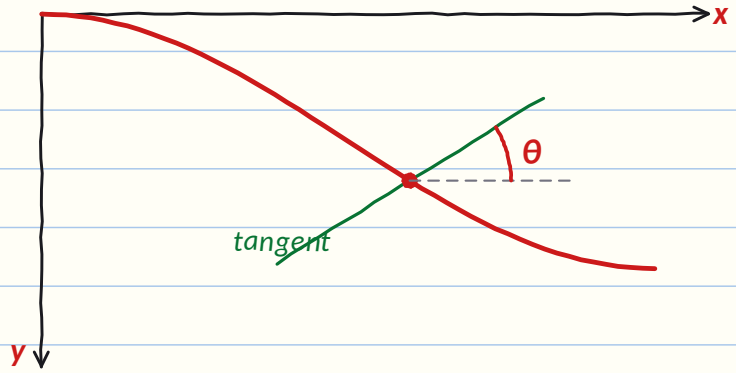
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4.1 Introduction to Deflection of Beams

- Under load a beam bends – the axis takes a curved shape called the **elastic curve** (deflection curve).
- Deflection (y) : vertical displacement of a point on the axis. Slope (θ) : angle between the tangent to the elastic curve and the original axis.
- Why? Serviceability limits : $\delta_{max} \le L/325$ (typ. L/360 for floors, L/180 for cantilevers) ; excess deflection cracks plaster, damages machines.

Assumptions

- Beam is initially straight, material homogeneous & elastic (E const.), small deflections (slope is small so tan θ ≈ θ).
- Deflection due to shear is neglected ; plane sections remain plane.



Slope $\theta = dy/dx$

Curvature $1/R \approx d^2y/dx^2$

From bending theory $M / I = E / R$

$E\,I\,d^2y/dx^2 = M$

Integrate once → slope

Integrate twice → deflection

Differential equation of the elastic curve

- From geometry : $1/R = (d^2y/dx^2)/([1 + (dy/dx)^2]^{3/2}) \approx d^2y/dx^2$ (slope small).
- Bending equation : $M/I = E/R \Rightarrow 1/R = M/EI$.

$$E\,I\,\frac{d^2y}{dx^2} = M$$

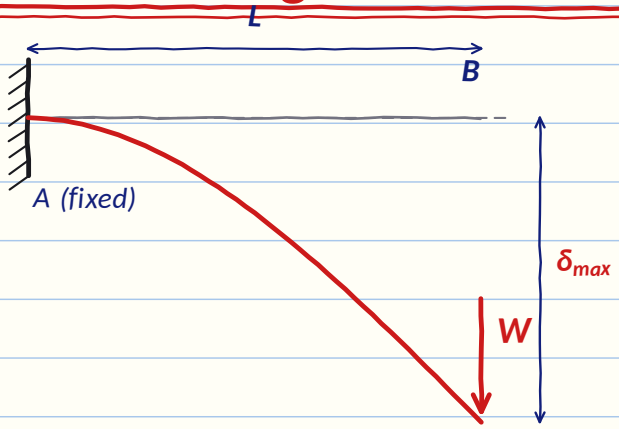
- Slope $\theta = dy/dx = 1/EI \int M\,dx + C_1$; deflection $y = 1/EI \iint M\,dx\,dx + C_1 x + C_2$.
- Sign : x → right, y ↑ +ve (so downward deflection is −ve), M sagging +ve. Constants C_1, C_2 from boundary conditions.

Support / point	Slope θ	Deflection y
Fixed end	0	0
Simple / roller support	≠ 0	0
Free end	≠ 0	≠ 0
Max deflection point	—	dy/dx = 0

Methods of finding slope and deflection

- (1) Double integration (2) Macaulay's method (3) Moment-area (Mohr's) method (4) Conjugate beam (5) Strain-energy (Castigliano).

4.2 Double Integration Method – Cantilevers



$M_x = -W(L - x)$ (x from fixed end A)

$EI\, d^2y/dx^2 = -W(L - x)$

slope & deflection at free end :

$\theta_B = W L^2 / 2EI$

$\delta_B = W L^3 / 3EI$

(a) Cantilever, point load W at free end B (fixed end A at x = 0)

- $M_x = -W(L - x) \Rightarrow EI\, y'' = -W(L - x)$
- $EI\, y' = -W(Lx - x^2/2) + C_1$; $y'(0) = 0 \Rightarrow C_1 = 0$
- $EI\, y = -W(Lx^2/2 - x^3/6) + C_2$; $y(0) = 0 \Rightarrow C_2 = 0$

$$\theta_B = \frac{W L^2}{2EI} \qquad \delta_B = \frac{W L^3}{3EI} \text{ (downward)}$$

(b) Cantilever with U.D.L. w over the whole span

- $M_x = -w(L - x)^2/2 \Rightarrow EI\, y'' = -w(L - x)^2/2$
- $EI\, y' = w(L - x)^3/6 + C_1$; $y'(0) = 0 \Rightarrow C_1 = -wL^3/6$
- $EI\, y = -w(L - x)^4/24 - (wL^3/6)x + C_2$; $y(0) = 0 \Rightarrow C_2 = wL^4/24$

$$\theta_B = \frac{w L^3}{6EI} \qquad \delta_B = \frac{w L^4}{8EI}$$

(c) Cantilever, load W at distance a from the fixed end

- Portion A–C bends ; portion C–B stays straight ($M = 0$) with slope θ_C.

$$\theta_C = \theta_B = \frac{W a^2}{2EI} \qquad \delta_C = \frac{W a^3}{3EI} \qquad \delta_B = \frac{W a^2}{6EI}(3L - a)$$

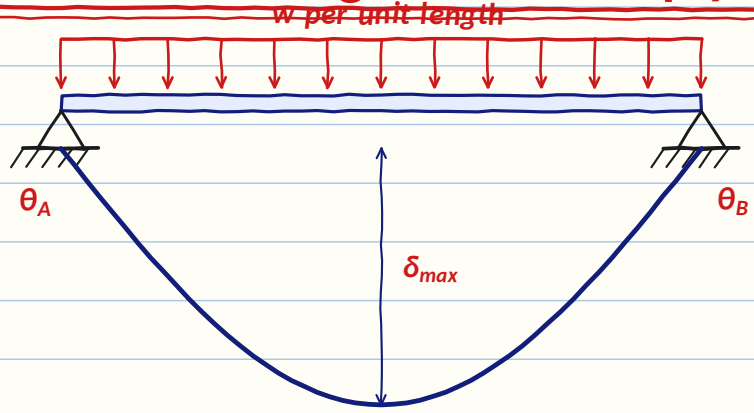
(d) Couple M at the free end

- $M_x = -M$ (constant) $\Rightarrow \theta_B = ML/EI$, $\delta_B = (ML^2)/2EI$.

(e) U.V.L. zero at free end, w at fixed end

- $M_x = - w (L - x)^3/6L \Rightarrow \theta_B = (w L^3)/24 E I$, $\delta_B = (w L^4)/30 E I$.

4.3 Double Integration – Simply Supported Beams



$M_x = w L x/2 - w x^2/2$

$\theta_A = -\theta_B = w L^3/24 E I$

$\delta_{max} = 5 w L^4/384 E I$ (at centre, $x = L/2$)

(a) Central point load W

- For $0 \le x \le L/2$: $M = W x/2 \Rightarrow E I y'' = W x/2$
- $E I y' = W x^2/4 + C_1$; slope = 0 at $x = L/2 \Rightarrow C_1 = - W L^2/16$
- $E I y = W x^3/12 - W L^2 x/16 + C_2$; $y(0) = 0 \Rightarrow C_2 = 0$

$$\theta_A = \theta_B = \frac{W L^2}{16 E I} \qquad \delta_{max} = \frac{W L^3}{48 E I} \text{ (at centre)}$$

(b) U.D.L. w over full span

- $M_x = w L x/2 - w x^2/2 \Rightarrow E I y' = w L x^2/4 - w x^3/6 + C_1$
- $E I y = w L x^3/12 - w x^4/24 + C_1 x + C_2$; $y(0)=0 \Rightarrow C_2 = 0$; $y(L)=0 \Rightarrow C_1 = - w L^3/24$

$$\theta_A = \theta_B = \frac{w L^3}{24 E I} \qquad \delta_{max} = \frac{5 w L^4}{384 E I}$$

(c) Eccentric point load W (a from A, b from B)

$$\delta_{under\ load} = \frac{W a^2 b^2}{3 E I L} \qquad \theta_A = \frac{W b (L^2 - b^2)}{6 L E I}$$

- Max deflection lies in the longer segment at $x = \sqrt{((L^2 - b^2)/3)}$ from A (for $a > b$).

(d) Couple M at end A (other end supported)

$$\theta_A = \frac{M L}{3 E I} \qquad \theta_B = \frac{M L}{6 E I} \qquad \delta_{max} = \frac{M L^2}{9\sqrt{3} E I} \text{ at } 0.577\ L$$

(e) U.V.L. (0 at A → w at B)

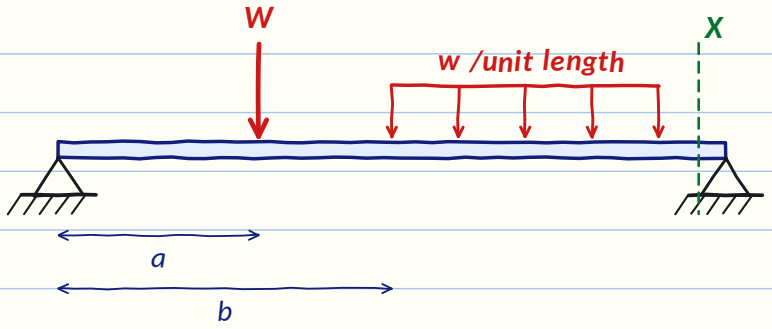
- $\theta_A = 7 w L^3/360 E I$; $\theta_B = w L^3/45 E I$; $\delta_{max} \approx 0.00652 w L^4/E I$ at 0.519 L from A.

Principle of superposition

When several loads act, slope/deflection = algebraic sum of the values due to each load separately (linear elastic, small deflection).

4.4 Macaulay's Method

Writes ONE bending-moment equation valid for the whole beam by using bracketed terms [x – a] that vanish when x < a. Integrate as a whole – constants C_1 and C_2 are common to all segments.



Take the section X–X in the LAST segment

Write M_x from the left end A

$M_x = R_A\, x - W\,[x-a] - w\,[x-b]^2/2$

Integrate everything as a whole;

[ ] term vanishes if x < its start

Rules

- Take the origin at the LEFT end ; write M_x for a section in the LAST segment (right-most).
- Point load W at a : – W [x – a] ; U.D.L. w beginning at b : – w $[x - b]^2/2$.
- A U.D.L. that stops before the end : extend it to the end and add an equal upward load + w $[x - c]^2/2$ from the stop point.
- Couple M_0 (clockwise) at d : + M_0 $[x - d]^0$.
- Integrate the brackets as $([x - a]^{n+1})/(n+1)$ – never expand them.
- A bracket whose value is negative (x < a) is dropped.

Example 1 : S.S. beam 6 m, W = 12 kN at 2 m from A (R_A = 8 kN)

$M_x = 8x - 12\,[x - 2]$

$E\,I\,y' = 4x^2 - 6\,[x - 2]^2 + C_1$

$E\,I\,y = (4/3)\,x^3 - 2\,[x - 2]^3 + C_1\,x + C_2$

$y(0) = 0 \Rightarrow C_2 = 0$; $y(6) = 0 \Rightarrow 288 - 128 + 6C_1 = 0 \Rightarrow$ **$C_1 = -26.67$**

Slope at A = – 26.67/EI ; deflection under load (x = 2) : E I y = 10.67 – 53.33 = **– 42.67 /EI kN·m³**

Max deflection where slope = 0 (x > 2) : $-2x^2 + 24x - 50.67 = 0 \Rightarrow x = 2.734$ m $\Rightarrow$ E I y = **– 46.45**

With E I = 10^4 kN·m² : $\delta_C = 4.27$ mm ; $\delta_{max} = 4.65$ mm ; $\theta_A = 0.00267$ rad

Example 2 : S.S. beam 6 m with U.D.L. 6 kN/m over the last 3 m (3 m ≤ x ≤

- Loads : total 18 kN at 4.5 m ⇒ R_B = 13.5 kN , R_A = 4.5 kN.

$M_x = 4.5\,x - 3\,[x-3]^2$

$E\,I\,y' = 2.25\,x^2 - [x-3]^3 + C_1$; $E\,I\,y = 0.375\,x^3 - 0.25\,[x-3]^4 + C_1\,x$

$y(6) = 0 \Rightarrow 81 - 20.25 + 6\,C_1 = 0 \Rightarrow$ **$C_1 = -10.125$**

$\theta_A = -10.125/EI$; $\theta_B = 2.25\times36 - 27 - 10.125 =$ **$+43.875/EI$**

Deflection at $x = 3$: $E\,I\,y = 10.125 - 30.375 =$ **$-20.25/EI$ kN·m^3**

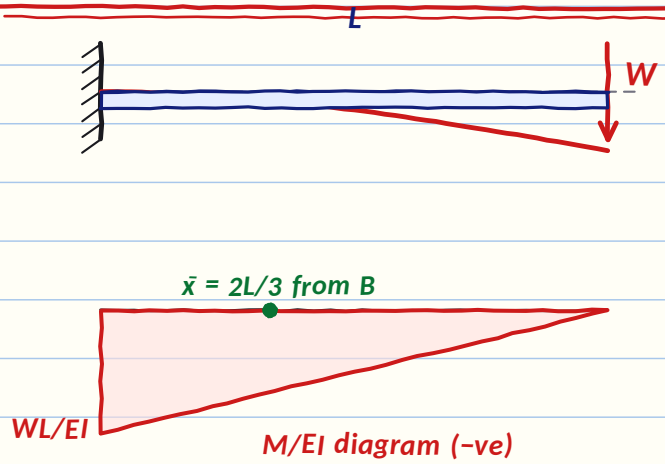
Overhanging beams

- Write M_x from the left end including both reactions ; boundary conditions : y = 0 at BOTH supports (not at the free end).
- Standard result – S.S. span L with overhang a carrying W at the free end :

$$\theta_B = \frac{W\,a\,L}{3\,E\,I} \qquad \theta_A = \frac{W\,a\,L}{6\,E\,I} \qquad y_C = \frac{W\,a^2(L+a)}{3\,E\,I} \quad \text{(overhang tip)}$$

- The portion between supports bends upward (hogging) – curvature reverses at the contra-flexure point.

4.5 Moment-Area Method (Mohr's Theorems)



Mohr's Theorem I :

$\theta_B - \theta_A$ = Area of M/EI diagram

Mohr's Theorem II :

$\delta = A\,\bar{x}$ (moment of area about B)

Here $A = \frac{1}{2} \cdot L \cdot WL/EI = W\,L^2/2EI$

$\delta_B = A \cdot \bar{x} = (W\,L^2/2EI)(2L/3) = W\,L^3/3EI$

Mohr's Theorem I : the change in slope between two points A and B of the elastic curve = area of the M/EI diagram between A and B.

$$\theta_B - \theta_A = \int_A^B \frac{M}{EI}\,dx = A_{M/EI}$$

Mohr's Theorem II : the deviation of point B from the tangent drawn at A = moment of the M/EI area between A and B about B.

$$\delta_{B/A} = A_{M/EI} \cdot \bar{x}_B$$

- $\bar{x}_B$ = distance of the centroid of the M/EI area from B. For a cantilever, tangent at the fixed end is horizontal ⇒ deviation = deflection.
- Useful areas : triangle ½ b h (centroid b/3 from the vertical side) ; parabola (vertex at the end) ⅓ b h (centroid 3b/4 from the vertex).

Example 3 : cantilever, W at free end

M/EI diagram – triangle, max = W L/EI at A, zero at B. Area = ½ × L × W L/EI = $W\,L^2/2EI$

θ_B = **$W\,L^2/2EI$** ; centroid 2L/3 from B ⇒ $\delta_B = (W\,L^2/2EI)(2L/3)$ = **$W\,L^3/3EI$**

Example 4 : cantilever with U.D.L.

$M/EI = w x^2/2EI$ (parabola, vertex at free end) ⇒ Area = ⅓ L ($w L^2/2EI$) = $w L^3/6EI$ = θ_B

centroid 3L/4 from B ⇒ $\delta_B = (w L^3/6EI)(3L/4) =$ **$w L^4/8EI$**

Example 5 : S.S. beam, central load

Tangent at C (midspan) is horizontal. θ_A = area of M/EI between A and C = ½ (L/2)(W L/4EI) = **$W L^2/16EI$**

δ_C = deviation of A from tangent at C = ($W L^2/16EI$)(L/3) = **$W L^3/48EI$**

4.6 Standard Results and Problems

Beam & loading	Max slope θ	Max deflection δ
Cantilever, W at free end	$W L^2/2EI$	$W L^3/3EI$
Cantilever, U.D.L. w	$w L^3/6EI$	$w L^4/8EI$
Cantilever, couple M at end	$M L/EI$	$M L^2/2EI$
Cantilever, U.V.L. (0 at tip)	$w L^3/24EI$	$w L^4/30EI$
S.S., W at centre	$W L^2/16EI$	$W L^3/48EI$
S.S., U.D.L. w	$w L^3/24EI$	$5 w L^4/384EI$
S.S., W at a, b	$W b (L^2-b^2)/6LEI$	$W a^2b^2/3EIL$
S.S., couple M at end	$M L/3EI$	$M L^2/9\sqrt{3}\ EI$

Example 6 (S.S. beam with U.D.L.)

$L = 4$ m, $w = 20$ kN/m, $E = 200$ GPa, $I = 1.2\times10^8$ mm^4.

$\delta_{max} = 5 w L^4/384 E I = 5\times20\times4000^4 / (384 \times 2\times10^5 \times 1.2\times10^8) =$ **2.78 mm**

Permissible $L/360 = 11.1$ mm ⇒ safe ✓ ; $\theta_A = w L^3/24EI =$

$20\times4000^3/(24\times2\times10^5\times1.2\times10^8) =$ **0.0022 rad**

Example 7 (Cantilever – superposition)

$L = 2$ m ; U.D.L. 5 kN/m over whole span + 8 kN at free end ; $E I = 4\times10^3$ kN·m^2.

$\delta_B = w L^4/8EI + W L^3/3EI = 80/32000 + 64/12000 = 0.0025 + 0.00533 =$ **7.83 mm**

Example 8 (Cantilever, moment-area)

$L = 2$ m, $W = 10$ kN at tip, $E I = 5\times10^3$ kN·m^2.

$\theta_B = W L^2/2EI = 10\times4/(2\times5000) =$ **0.004 rad** ; $\delta_B = W L^3/3EI = 80/15000 =$ **5.33 mm**

Example 9 (Cantilever, load not at the tip)

$L = 3$ m, $W = 6$ kN at $a = 1$ m from fixed end, $E I = 4\times10^3$ kN·m^2.

$\theta_C = \theta_B = W a^2/2EI = 6\times1/(2\times4000) =$ **0.00075 rad**

$\delta_C = W a^3/3EI = 6/12000 =$ **0.5 mm** ; $\delta_B = W a^2(3L - a)/6EI = 6\times8/24000 =$ **2.0 mm**

Example 10 (Macaulay with a couple)

S.S. beam 4 m, clockwise couple 8 kN·m at 1 m from A ⇒ $R_B = 2$ kN ↑, $R_A = 2$ kN ↓.

$M_x = -2x + 8\,[x - 1]^0$

$E I y' = -x^2 + 8\,[x - 1] + C_1$; $E I y = -x^3/3 + 4\,[x - 1]^2 + C_1 x$

$y(4) = 0 \Rightarrow -21.33 + 36 + 4 C_1 = 0 \Rightarrow$ **$C_1 = -3.667$**

$\theta_A = -3.667/EI$; deflection at the couple ($x = 1$) : $E I y = -0.333 - 3.667 =$ **-4.0**

/EI

Example 11 (Overhanging beam, standard result)

S.S. span $L = 4$ m, overhang $a = 1.5$ m, $W = 10$ kN at the tip, $EI = 2\times10^4$ kN·m^2.

$\theta_B = W a L/3EI = 10\times1.5\times4/(6\times10^4) =$ **0.001 rad**

$y_C = W a^2(L + a)/3EI = 10\times2.25\times5.5/(6\times10^4) =$ **2.06 mm** (downward at the tip)

Example 12 (Cantilever with U.D.L. – moment-area check)

$L = 2$ m, $w = 5$ kN/m, $EI = 4\times10^3$ kN·m^2.

$\theta_B = w L^3/6EI = 5\times8/24000 =$ **0.00167 rad** ; $\delta_B = w L^4/8EI = 80/32000 =$ **2.5 mm**

Steps to solve any deflection problem

- 1. Draw the free-body diagram ; find reactions.
- 2. Write M_x (Macaulay form if loads are not over the full span).
- 3. Integrate twice ; keep the constants C_1, C_2.
- 4. Apply boundary conditions (supports : $y = 0$; fixed end : $\theta = 0$).
- 5. Substitute x to get slope/deflection ; for y_{max} put $dy/dx = 0$.
- 6. Convert units : EI in kN·m^2 with lengths in m gives answer in m.

Short questions

- State Mohr's theorems. A: see 4.5 – slope change = M/EI area ; deviation = moment of M/EI area.
- What is the Macaulay bracket? A: $[x - a]^n = 0$ for $x < a$.
- Where is the deflection maximum? A: where slope $dy/dx = 0$.
- Boundary conditions for fixed end ? A: $\theta = 0$, $y = 0$.